

Introduction to Power BI, Charts, DAX & Creating Reports

Question 1:

Define Power BI and What are the key components of the Power BI ecosystem? Briefly explain:

- **Power BI Desktop**
- **Power BI Service**
- **Power BI Mobile**
- **Power BI Gateway**

Answer:

Power BI is a business analytics tool developed by Microsoft that helps users visualize data, create interactive dashboards, and generate insights for better decision-making. It allows you to connect to multiple data sources, transform data, and share reports easily.

Key Components of Power BI Ecosystem

1. Power BI Desktop

Power BI Desktop is a free application installed on a computer. It is mainly used for:

- Connecting to different data sources (Excel, SQL, etc.)
- Cleaning and transforming data (using Power Query)
- Creating data models and relationships
- Designing reports and visualizations

It is the main development tool where reports are created.

2. Power BI Service

Power BI Service is a cloud-based platform (online service). It is used for:

- Publishing reports created in Power BI Desktop
- Sharing dashboards with others
- Collaborating with team members
- Accessing reports from anywhere via internet

It helps in real-time data sharing and collaboration.

3. Power BI Mobile

Power BI Mobile is an app available on smartphones and tablets. It allows users to:

- View dashboards and reports on mobile devices
- Get real-time alerts and notifications
- Stay updated with business insights anytime, anywhere

Useful for managers and professionals who need data on the go.

4. Power BI Gateway

Power BI Gateway is a bridge between on-premises data sources and Power BI Service. It:

- Enables secure data transfer from local systems to the cloud
- Keeps data updated automatically (scheduled refresh)
- Ensures data privacy and security

Important for organizations using local databases.

Question 2:

Compare the following Power BI visuals:

● Pie Chart vs Donut Chart

● Bar Chart vs Column Chart When would you prefer one over the other? Give one example for each pair.

Answers:

1. Pie Chart vs Donut Chart

Pie Chart:

- Displays data as slices of a circle
- Shows proportion or percentage of each category
- Best for a small number of categories

Donut Chart:

- Similar to pie chart but with a hole in the centre
- More visually appealing and modern
- Can display additional information in the centre

When to use:

- Use Pie Chart when you want a simple and clear representation of proportions

- Use Donut Chart when you want a better visual design or need to show extra information in the centre.

Example:

- Showing market share of companies → Pie Chart
- Showing sales distribution with total value in centre → Donut Chart

2. Bar Chart vs Column Chart

Bar Chart:

- Displays data horizontally
- Categories are on the vertical axis
- Better for long category names

Column Chart:

- Displays data vertically
- Categories are on the horizontal axis
- Best for comparing values over time

When to use:

- Use Bar Chart when category names are long or you have many categories
- Use Column Chart when showing trends over time (monthly, yearly data)

Example:

- Comparing sales by product names (long names) → Bar Chart
 - Showing monthly revenue growth → Column Chart
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Question 3:

Explain the significance of:

- **Star schema vs Snowflake schema**
- **Primary key vs foreign key in relationships (Power BI) Why is cardinality important?**

Answers:

Data Modeling Concepts in Power BI

1. Star Schema vs Snowflake Schema

Star Schema:

- Consists of one central fact table connected to multiple dimension tables
- Dimension tables are not normalized (no further splitting)

- Simple structure and easy to understand
- Provides faster query performance

Snowflake Schema:

- Extension of star schema where dimension tables are further normalized into multiple related tables
- More complex structure
- Reduces data redundancy
- Slightly slower performance due to multiple joins

Significance:

- Star schema is preferred in Power BI because it is simple and improves performance
- Snowflake schema is used when data needs to be highly structured and normalized

2. Primary Key vs Foreign Key (in Power BI Relationships)

Primary Key:

- A column that uniquely identifies each record in a table
- Contains unique and non-null values

Example: Customer ID in Customer table

Foreign Key:

- A column in one table that refers to the primary key of another table
- Used to create relationships between tables
- Example: Customer ID in Sales table

Significance:

- Helps in connecting tables and building relationships in Power BI
- Ensures accurate data modeling and prevents duplication

Why is Cardinality Important?

Cardinality defines the type of relationship between two tables. Common types:

- One-to-One (1:1)
- One-to-Many (1:M)
- Many-to-Many (M:M)

Importance:

- Determines how data is filtered and connected between tables
- Affects accuracy of reports and calculations

- Helps avoid issues like duplicate or incorrect data
- Improves performance and data integrity

In Power BI, correct cardinality ensures that visuals show accurate results when combining data from multiple tables.

Question 4 :

Differentiate between:

- **Calculated column vs Measure Also, define Row context and Filter context with simple examples.**

Answer:

1. Calculated Column vs Measure

Calculated Column:

- Created inside a table and calculated for each row
- Stored in the data model (takes memory)
- Used for row-level calculations
- Can be used in slicers, filters, and axes

Measure:

- Created using DAX and calculated dynamically
- Not stored in the table (calculated at runtime)
- Used for aggregations (SUM, AVG, COUNT, etc.)
- Mainly used in visuals

Difference:

- Calculated column works at **row level**, while measure works at **aggregate level**
- Calculated column increases model size, measure does not

Example:

- Calculated Column → Profit = Sales - Cost (for each row)
- Measure → Total Profit = SUM(Profit)

2. Row Context

Definition:

Row context means the calculation is performed **row by row** in a table.

Example:

If you create a calculated column:

Profit = Sales - Cost

Power BI calculates this for each individual row separately.

3. Filter Context**Definition:**

Filter context means calculations change based on filters applied (like slicers, visuals, or conditions).

Example:

If you create a measure:

Total Sales = SUM(Sales)

When you apply a filter (e.g., Year = 2024), the measure will only calculate sales for 2024.

- **Row Context** → Works on individual rows
 - **Filter Context** → Works based on applied filters
 - **Calculated Column** → Static, stored, row-wise
 - **Measure** → Dynamic, calculated on demand
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Question 5:**What is the difference between a report and a dashboard in Power BI?****Answer:**

In Microsoft Power BI, a **report** and a **dashboard** are different in purpose and structure.

Report

- A report is a **detailed collection of visuals** such as charts, tables, graphs, and slicers.
- It can have **multiple pages**.
- Reports are mainly used for **deep analysis and data exploration**.
- Users can interact with filters, drill-throughs, and detailed data.
- Reports are usually created in **Power BI Desktop**.

Example:

A sales report containing separate pages for revenue, profit, region-wise sales, and customer analysis.

Dashboard

- A dashboard is a **single-page summary view** of important metrics.
- It displays key visuals pinned from one or multiple reports.
- Dashboards are used for **quick monitoring and decision-making**.

- They provide a high-level overview instead of detailed analysis.
- Dashboards are mainly created in **Power BI Service**.

Example:

A dashboard showing total sales, profit, top-performing region, and monthly growth in one screen.

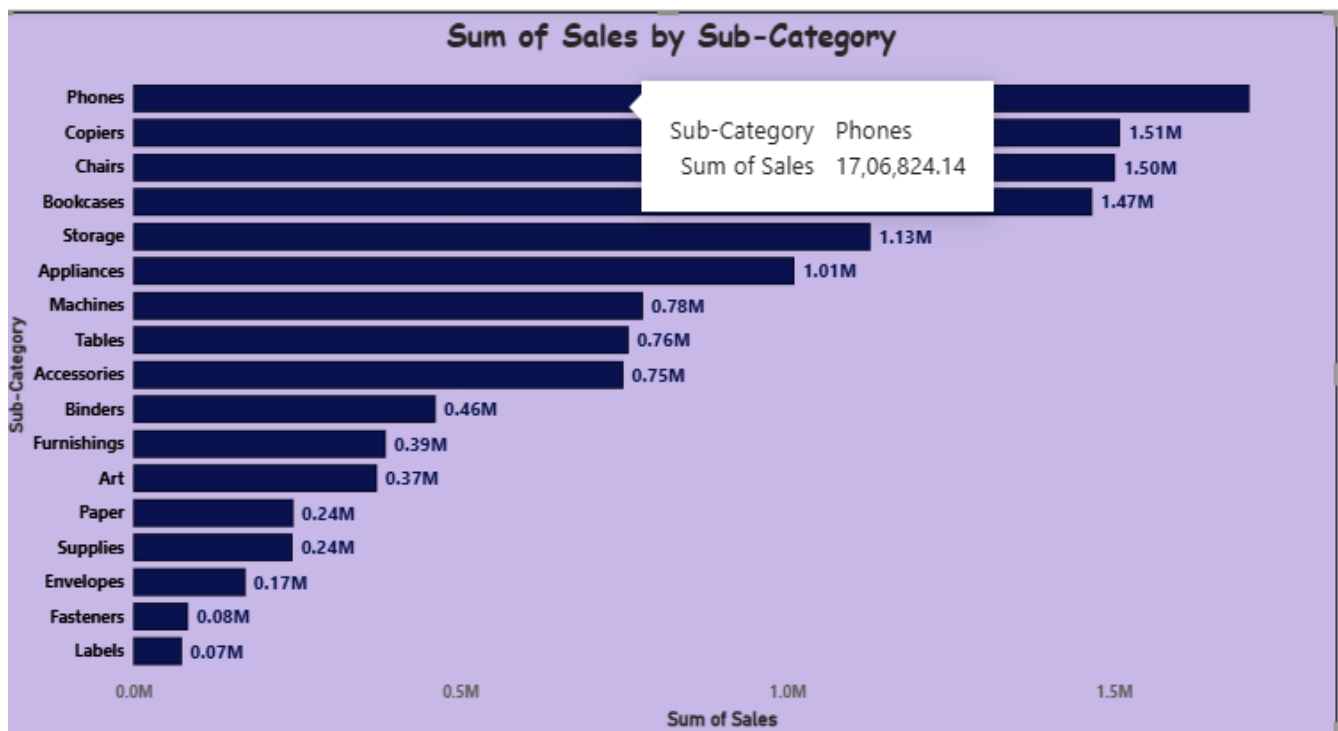
Main Difference

- Report = Detailed Analysis (multiple pages)
- Dashboard = Quick Summary (single page)

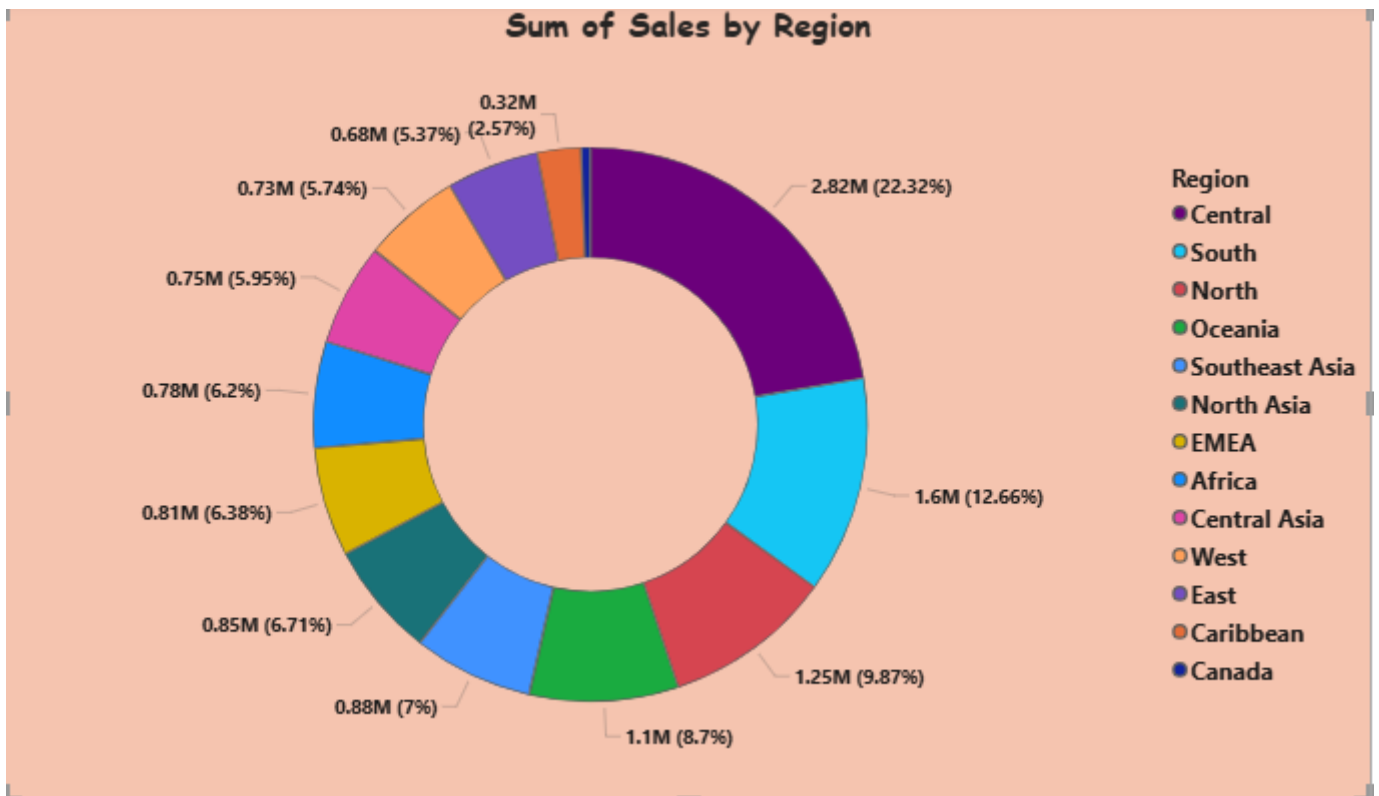
Question 6 :

Using the Sample Superstore dataset:

- Create a Clustered Bar Chart to display Total Sales by Sub-Category



- Create a Donut Chart for Sales % by Region Provide screenshots of both visuals.



Question 7 : Write and apply the following measures:

- Total Profit = SUM([Profit])
- Average Discount = AVERAGE([Discount]) Display both in a KPI Card, and use a Line Chart to show profit trend over months. Add visuals and DAX formulas.



Question 8 : Implement a DAX measure that calculates the percentage of total sales by product category.

Percent of Total sales = VAR TotalSelected =
CALCULATE(SUM(Global_Superstore2[Sales]),Global_Superstore2[Sub-Category] IN {
 "Electronics", "Clothing", "Home Appliances", "Books",
 "Tables & Chairs", "Toy", "Sports Equipment", "Office Supplies",
 "Beauty Products", "Garden Supplies", "Jewelry", "Automotive"
}) RETURN DIVIDE(SUM(Global_Superstore2[Sales]),TotalSelected)

Question 9 : • Create a DAX Measure for Total Profit

- **Use it in a Waterfall Chart to analyze how different Sub-Categories contribute to overall profit**
- **Add a Slicer for Region to filter the visual**
- **Write brief business insights (4–5 lines) from the chart and provide 2–3 data-driven recommendations to improve profit. Provide a steps, screenshot of the Waterfall chart and the DAX formula**

In Microsoft Power BI, complete the task using the following steps.

Step 1: Create DAX Measure for Total Profit

Go to **Modeling** → **New Measure** and write:

Total Profit = SUM('Sample Superstore'[Profit])

Step 2: Create Waterfall Chart

Steps

1. Select the **Waterfall Chart** visual.
2. Drag:
 - **Sub-Category** → Category
 - **Total Profit** → Y-axis

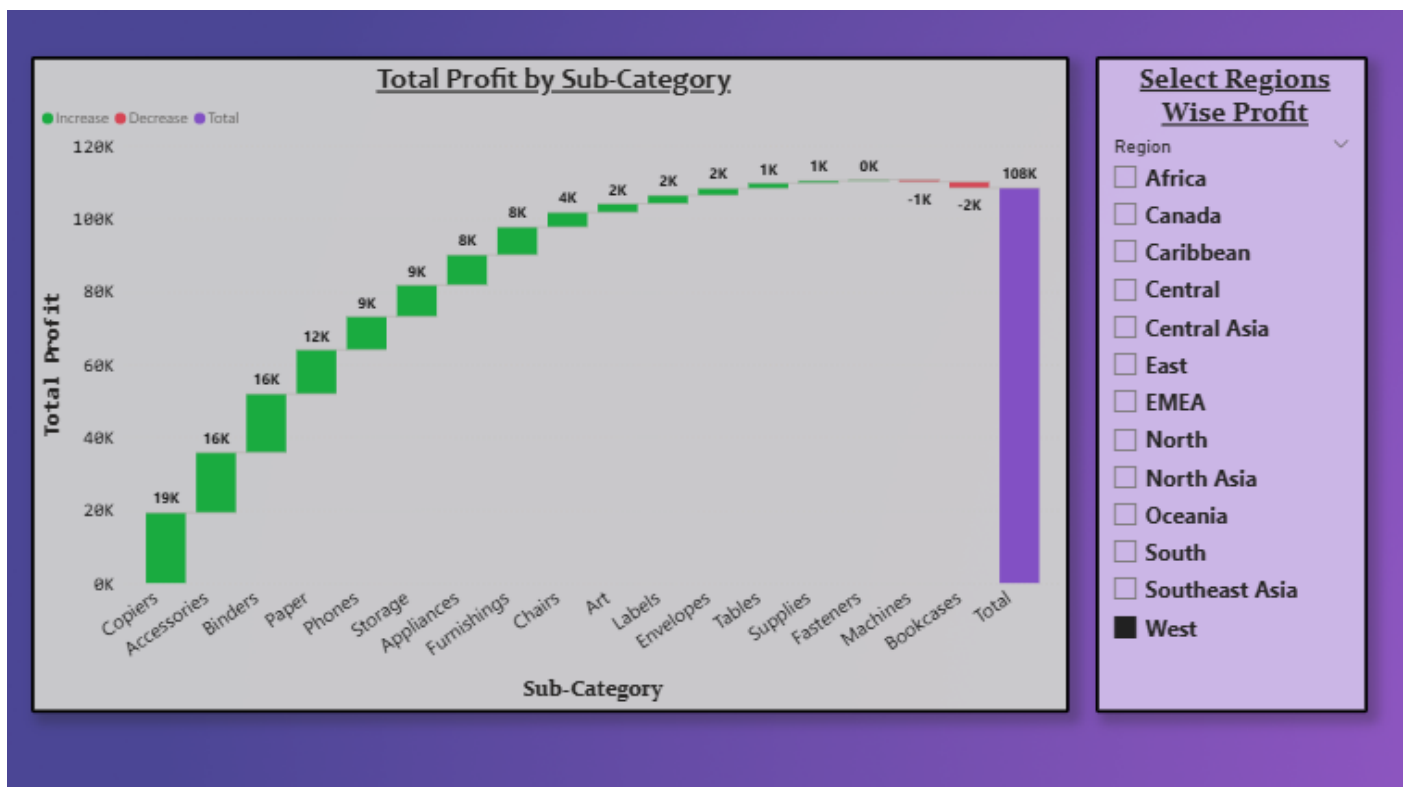
The chart will show:

- Which sub-categories contribute positively to profit
- Which sub-categories reduce overall profit

Step 3: Add Region Slicer

Steps

1. Select the **Slicer** visual.
2. Drag **Region** into the slicer field.
3. Users can now filter the Waterfall Chart by:
 - East
 - West
 - Central
 - South



Business Insights

1. Some sub-categories contribute significantly to overall profit, while a few generate losses.
2. High-profit categories indicate strong product demand and better margins.
3. Loss-making categories may be affected by heavy discounts or low sales volume.
4. Profit performance changes across different regions when filtered using the slicer.
5. The business should focus more on consistently profitable product categories.

Data-Driven Recommendations

1. Reduce discounts on low-profit or loss-making sub-categories to improve margins.
 2. Increase inventory and marketing efforts for high-performing products like Phones or Copiers.
 3. Analyze regional sales patterns and focus on regions generating higher profitability.
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Question 10 :

Scenario:

VitaTrack Wellness, a digital health company in FitZone, has collected data on users' daily habits and health vitals. The analytics team is tasked with drawing actionable insights from this data to improve lifestyle suggestions and prevent heart-related risks.

Your Task:

Using the provided dataset (includes Age, Gender, BMI, Steps, Calories, Sleep, Heart Rate, Blood Pressure, Smoking, Alcohol, Exercise, Diabetic & Heart Disease status):

Build a one-page Power BI dashboard that answers:

1. Are users maintaining a balanced lifestyle (Steps, Sleep, Calories)
2. What lifestyle patterns (Smoking, Alcohol, BMI, etc.) indicate heart disease risk?
3. Is there any visible relationship between Sleep and Physical Activity?
4. How does BMI vary across Age Groups and Genders?
5. What is the impact of smoking and alcohol on heart rate and blood pressure?
6. Segment people based on their health activity to suggest lifestyle changes

VitaTrack Wellness Dashboard

10.72K

Average Steps

6.91

Average Sleep

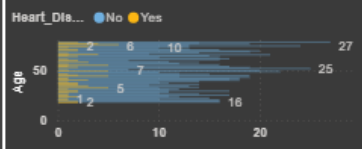
2.33K

Average Calories

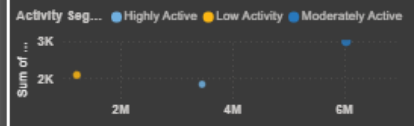
93

Heart Disease Count

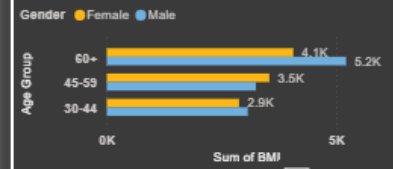
Count of Smoker by Age and Heart_Disease



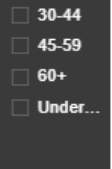
Sum of Daily_Steps, Sum of Hours_of_Sleep and Sum of Calories_Intake by Activity Segment



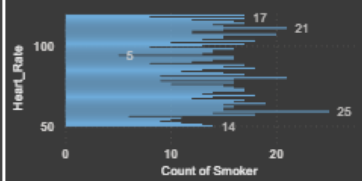
Sum of BMI by Age Group and Gender



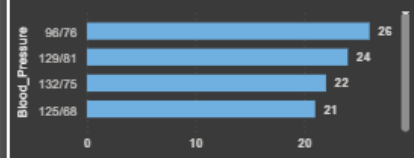
Age Gr...



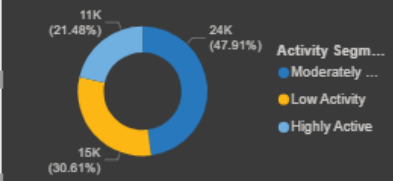
Count of Smoker by Heart_Rate



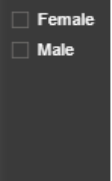
Sum of Alcohol_Consumption_per_Week by Blood_Pressure



Sum of Age by Activity Segment



Gen...



Final Business Insights

1. Users with higher daily steps usually maintain better sleep patterns.
2. Smokers and alcohol consumers show comparatively higher heart rate levels.
3. Higher BMI is more common in middle-aged groups.
4. Low activity users may face greater lifestyle-related health risks.
5. Active users generally show healthier wellness indicators.

Recommendations

1. Encourage low-activity users to improve daily physical movement.
2. Promote awareness programs for smoking and alcohol reduction.
3. Suggest personalized health plans based on activity segments.
4. Improve sleep tracking and wellness reminders for users.